

Causal Disentanglement of User-Induced and Assigned Persona in Language Models

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1 Motivation

Large Language Models (LLMs) are often deployed in conversational settings where they are instructed to have a certain persona. However, during multi turn interactions, models often exhibit *personality mirroring*, a tendency to absorb and reflect the stylistic traits of the user (Sprejer et al., 2026). Additionally across long context conversations, models suffer from *persona drift*, where a system prompted behavior constraint gradually degrades (Lu, 2026). Though these processes have been documented, their underlying representational mechanisms are not explored. It is unclear whether persona drift arises from the gradual decay of prompt behavior, or from specific internal circuits routing the user’s context into the model’s latent state. Understanding this distinction remains an open question in mechanistic interpretability.

To answer this question, we must isolate the two latent variables involved: persona assigned through system prompt and the persona induced by the user through the conversation. As we discuss below, this separation is non trivial for standard probing methods, which leads to our circuit - level approach.

2 Related Work

Circuit Localization and Role Dissolution. Two recent findings provide the mechanistic foundation for our inquiry. First, Izawa (2026) demonstrate that stylistic modulation is not broadly distributed across the residual stream, but is concentrated within a sparse subset of attention heads—termed Style Modulation Heads (SMH)—such as Layer 20, Heads 3/5/28 in Qwen2.5-7B. Second, Anonymous (2026) reveal that explicit conversational role boundaries (e.g., User: versus Assistant:) dissolve within latent representations, where stylistic geometry dominates structural tags. This explains why naive linear probes

trained across conversation positions suffer from severe syntactic confounding. We build directly upon these insights by investigating whether the SMH acts as the causal circuit responsible for routing user-induced style across dissolved role boundaries.

3 Proposed Methodology

Our approach consists of three stages: a controlled counterfactual setup, latent-space analysis, and targeted interventions at the identified SMHs.

3.1 Counterfactual Setup

We first construct conversations where the user’s intent and the assistant’s response are held fixed while the user’s stylistic expression is varied. For example, we word the same request in formal, impolite, aggressive or neutral language. Then we teacher - force the fixed assistant response across all conditions and then check activations for the assistant token positions within the SMHs. This will enable us to study how changes in the user style lead to changes in the assistant’s internal representations.

3.2 Latent space analysis

We then vary both the assigned persona and the user expressed style. Let Z denote the collected SMH activations, X_A the assigned persona, and X_U the user-induced style. We model the activations using a linear regression framework of the form:

$$Z = \beta_0 + \beta_A X_A + \beta_U X_U + \beta_I (X_A \otimes X_U) + \epsilon. \quad (1)$$

where β_0 is the baseline activation, β_A and β_U capture the effects of the assigned persona and user style respectively, β_I captures their interaction, and ϵ represents residual variation. We use the corresponding coefficients to estimate activation subspaces A and U associated with the assigned and user-induced personas.

Then , we compare these subspaces using principal angles to determine whether there is any overlap between these two sources of stylistic influence. If there is a large overlap then that suggests that both of these are entangled in the internal representation without a clean separation.

3.3 Targeted Interventions

Finally we will test our hypothesis during targeted runtime changes. Specifically, during autoregressive generation, we intervene at the localized SMHs by projecting activations away from the estimated U subspace obtained above. We will evaluate whether this actually reduced user-style mirroring and multi turn persona drift while preserving the assistant’s assigned persona.

Together , these experiments will provide us with a mechanistic account of both kinds of persona drift , and test whether it can be selectively controlled through targeted interventions.

References

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